

HR Attrition Analysis and Employee Retention Strategy

A Data-Driven Business Analytics Case Study



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EXECUTIVE SUMMARY

This project analyses employee attrition using the IBM HR Analytics dataset to identify the key factors influencing workforce turnover. An interactive Power BI dashboard was developed to examine attrition trends across departments, job roles, compensation, employee satisfaction, work-life balance, overtime, and career growth factors.

The analysis found an attrition rate of approximately 16.1%, with higher turnover observed among employees working overtime, employees in lower salary bands, and those with lower job satisfaction and delayed promotions. The findings highlight critical employee segments at greater risk of leaving the organization.

The study provides data-driven insights to support HR decision-making and recommends focusing on employee engagement, career development, workload management, and compensation strategies to improve retention and workforce stability.

BUSINESS PROBLEM

High Employee Attrition is impacting Productivity, Hiring Costs and Workforce Stability

1470

Total Employees

237

No. of Employees
Left

16.1%

Attrition Rate

KEY CHALLENGES

1. **High Hiring Costs**
2. **Low Productivity**
3. **Employee Turnover**
4. **Knowledge Drain**

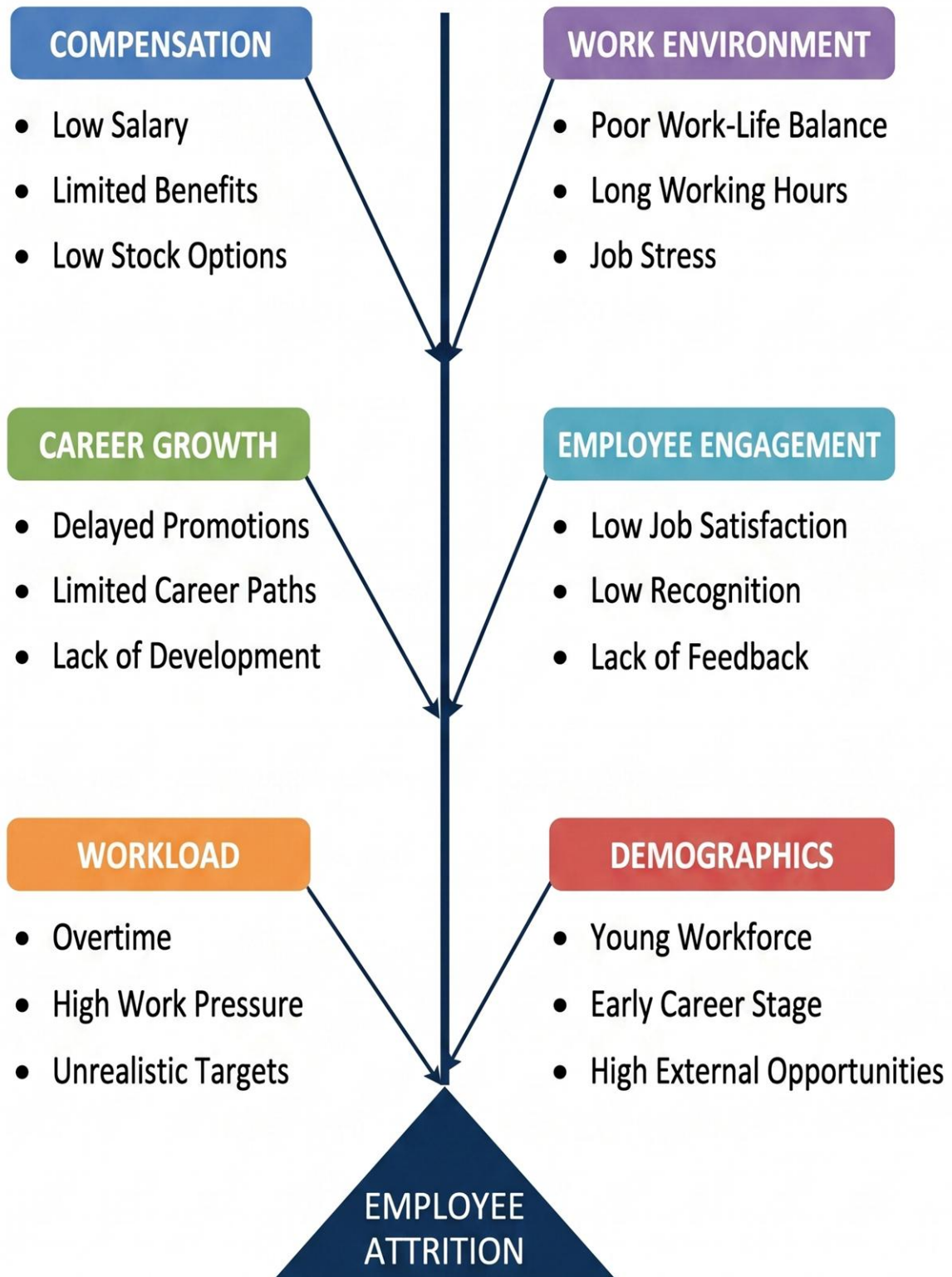
BUSINESS QUESTIONS

- Who is leaving?
- Why are they leaving?
- Which departments are most affected?
- What actions can improve retention?

KEY FINDINGS

- **Sales Department** has the highest attrition.
- Employees working **overtime** are more likely to leave.
- Low **salary band** employees show higher attrition.
- Low **job satisfaction** is strongly linked to attrition.
- Employees with long **promotion delays** are at higher risk.

ROOT CAUSE ANALYSIS



HIGH RISK EMPLOYEE PROFILE

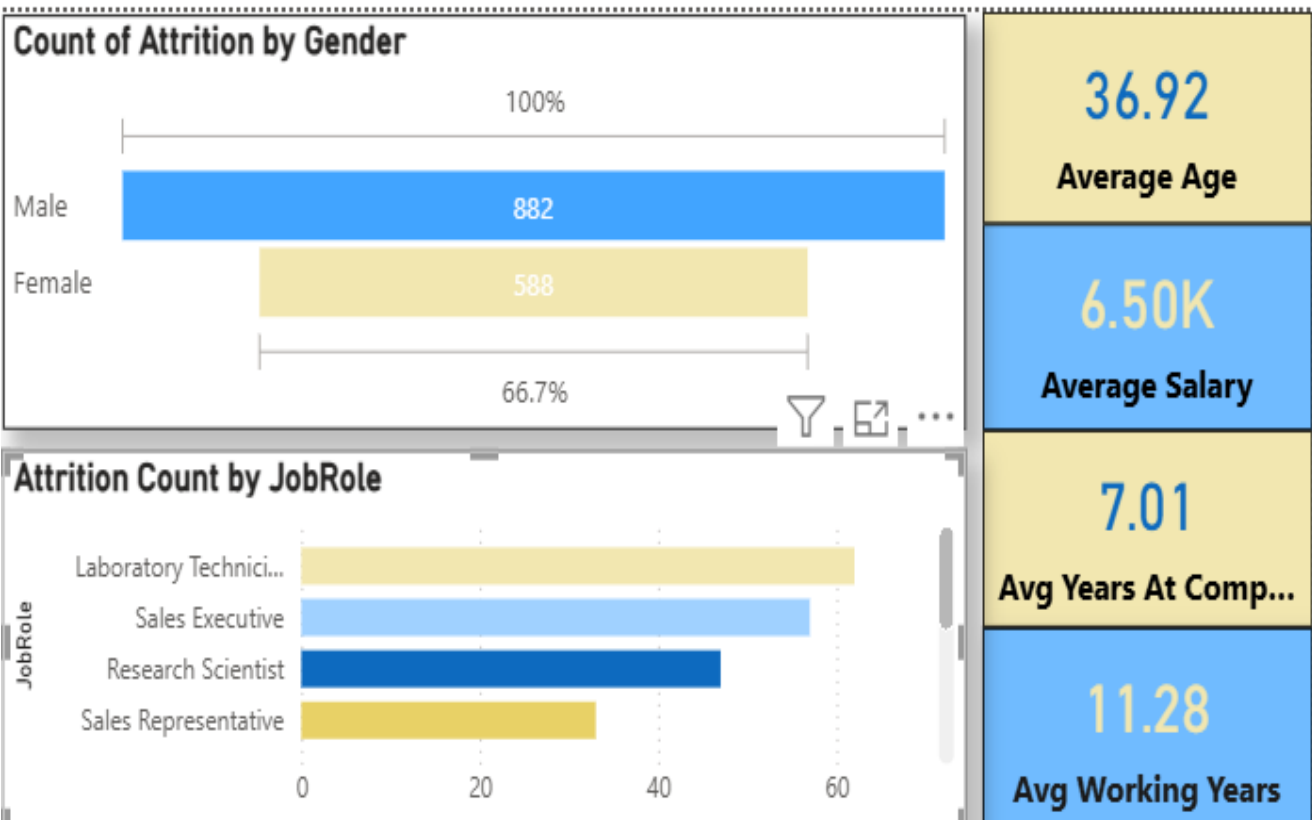
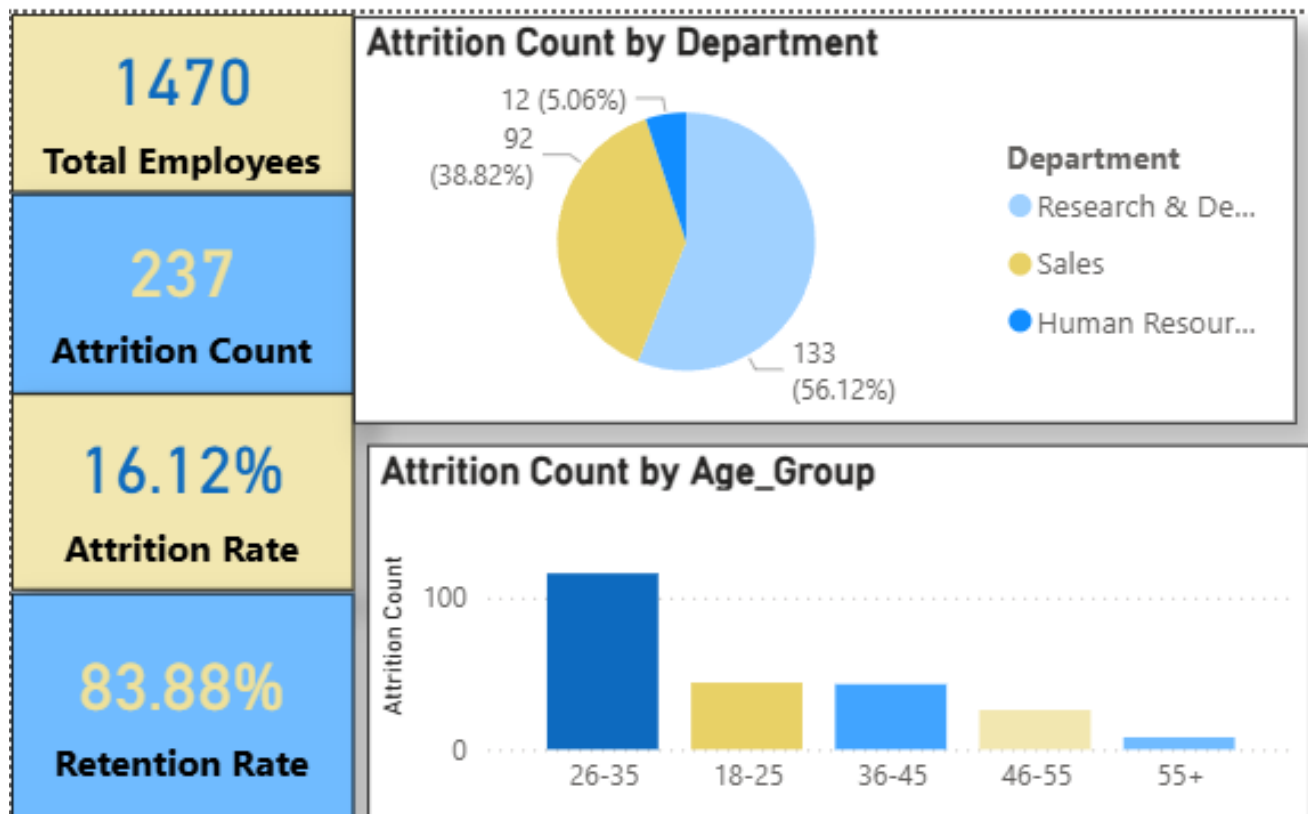


 Age Group	26 – 35 Years
 Department	Sales
 Job Role	Sales Executive / Laboratory Technician
 Salary Band	Low
 Overtime	Yes
 Years at Company	1 – 3 Years
 Promotion Risk	High (No Promotion in 2+ Years)
 Job Satisfaction	Low (1 – 2)



Employees with these characteristics are 2–3x more likely to leave the organization.

DASHBOARD SNAPSHOT



RECOMMENDATIONS

Improve Career Growth

- Create clear career paths and promotion guidelines.
- Implement regular performance & career discussions.

Reduce Overtime & Workload

- Optimize resource allocation and project planning.
- Monitor overtime and prevent burnout.

Review Compensation Strategy

- Benchmark salaries with market standards.
- Improve variable pay and reward top performers.

Strengthen Employee Engagement

- Conduct regular engagement surveys.
- Recognize achievements and encourage feedback.

EXPECTED BUSINESS IMPACT

CURRENT STATE

16.1%

Attrition Rate



TARGET STATE

12%

Attrition Rate

BENEFITS

- ✓ Reduced Hiring Costs
- ✓ Better Productivity
- ✓ Improved Employee Retention
- ✓ Stronger Workforce Stability
- ✓ Enhanced Organizational Culture



By addressing key attrition drivers, the organization can achieve sustainable growth through a stable and engaged workforce.

CONCLUSIONS

- * The analysis identified overtime, compensation, job satisfaction, work-life balance, and career growth as major drivers of employee attrition.
- * Certain departments, job roles, and employee segments showed higher turnover risk.
- * Data-driven retention strategies can help reduce attrition and improve workforce stability.
- * Focusing on employee engagement, career progression, and workload management can enhance retention and organizational performance.